

Labeling, Targets, and Prediction Horizons

Define causal outcomes and control leakage across overlapping prediction horizons.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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How to use this manual

INTERMEDIATE

1. BEGIN WITH THE DECISION

Name who acts, at what cutoff, with what latency, and on which instrument. A target is useful only when its entry reference and outcome window describe a feasible decision.

2. READ EVERY CLOCK

Separate feature availability, decision, order eligibility, entry, exit, label maturity, and revision time. A chronological dataset can still leak when these clocks are collapsed.

3. MATCH THE OUTCOME TYPE

Choose endpoint return, relative return, rank, barrier, excursion, volatility, or duration because it matches the strategy mechanics. Do not choose a label solely because it produces a better metric.

4. INSPECT INTERVAL GEOMETRY

Draw each label as an interval. Overlap, purging, embargo, group structure, and effective sample size follow from those intervals and cannot be repaired by random splitting.

5. TREAT LABELS AS SEARCHED OBJECTS

Horizons, thresholds, benchmarks, scales, barriers, and sampling rules are researcher choices. Select them inside training or inner validation and disclose the full family.

6. KEEP RAW OUTCOMES

Preserve raw returns, prices, paths, quality states, and alternative definitions beside transformed targets. This lets reviewers distinguish economics from the selected statistical encoding.

7. GOVERN MATURITY AND REVISIONS

Score only mature cohorts, label provisional results, freeze experiment artifacts, and create new versions for corrected history. Never silently overwrite the evidence behind a model decision.

DECISION STANDARD

Promote a target only when its meaning, causality, coverage, noise, overlap, metric, policy link, lineage, maturity process, and monitoring actions are defensible.

Notation and target-state contract

INTERMEDIATE

SYMBOL	OBJECT	MEANING
t_d	decision time	Cutoff when the forecast becomes actionable
t_e	entry time	Modeled executable start of economic exposure
t_x	exit time	End of the outcome or realized exit event
q	prediction horizon	Declared future span measured in time or events
$y_{(i,t)}$	target value	Outcome for instrument i and prediction decision t
$R_{(s,e)}$	forward return	Total return from entry s to exit e
U_t	eligible universe	Point-in-time peer set used by relative labels
M_t	maturity state	Provisional, mature, revised, censored, or excluded
$I_t=(s,e]$	label interval	Future events consumed by the target
w_i	sample weight	Declared influence in fitting or evaluation

TARGET BUNDLE

Publish value, raw outcome, unit, entry and exit times, horizon, benchmark, maturity, eligibility, censor state, quality reason, source vintage, label version, and lineage. The scalar alone is not sufficient for audit or validation.

CAUSAL INEQUALITY

Features and fitted parameters stop at the decision cutoff. The modeled entry follows the decision, the target consumes only future outcome events, and scoring waits until those events mature under the contract.

Target design and operation workflow

INTERMEDIATE

1

State the decision

Actor, cutoff, latency, side, instrument, and utility

2

Choose the outcome

Return, rank, risk, barrier, path, event, or duration

3

Define clocks

Feature cutoff, entry, interval, exit, maturity, and revision

4

Set conventions

Prices, adjustments, units, benchmark, costs, and universe

5

Handle incomplete cases

Eligibility, censoring, delistings, halts, and ambiguity

6

Map interval geometry

Overlap, groups, purge, embargo, and effective evidence

7

Validate target family

Nested selection, uncertainty, calibration, and policy value

8

Materialize artifact

Source snapshot, lineage, tests, maturity, and version hash

9

Monitor feedback

Cohorts, coverage, revisions, prevalence, drift, and utility

10

Govern changes

Parallel rebuild, approval, migration, incident, and rollback

The target contract

INTERMEDIATE

ONE-SENTENCE MEANING

A target contract defines the future outcome a model is asked to estimate, including its clock, start, end, units, population, maturity rule, and data vintage. It makes the dependent variable reproducible instead of leaving it as an ambiguous shifted column.

MEMORY JOGGER

Write the decision first, then the target that would inform it. A complete definition names the prediction timestamp, execution assumption, outcome window, benchmark, costs, missing policy, and versioned source.

WHY IT MATTERS IN QUANT RESEARCH

Small changes in label endpoints can change economic meaning, sample overlap, class balance, and apparent skill. Without a contract, two experiments can report the same metric while predicting different objects.

FORMULA INTUITION

$y_{-}(i,t) = L(D \text{ available after decision } t \text{ over interval } (s_{-}t, e_{-}t]; \text{ contract } v)$. The feature cutoff is at or before t , while the outcome must begin after the assumed action can occur.

SIMPLE MARKET EXAMPLE

A field named `forward_return_5` is incomplete until it says whether the trade enters at today's close or next open, exits five sessions later, includes dividends, and subtracts a benchmark or trading cost. Each choice changes the realized target.

PYTHON / SYSTEM IMPLEMENTATION

Store label ID, semantic version, decision calendar, entry and exit rules, price fields, adjustment policy, horizon, benchmark, cost model, maturity state, and lineage. Unit tests should reconstruct selected rows directly from source events.

CONFUSIONS TO AVOID

A dataframe shift is not a contract because row order may hide missing sessions, delistings, or asynchronous assets. Renaming a column while retaining changed semantics also creates false comparability across experiments.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label registry feeds training-set builders, validation splitters, metric selection, and post-trade attribution. Model artifacts bind to an immutable label version so performance remains interpretable after data revisions.

RELATED CONCEPTS AND QUICK RECALL

Related: dependent variable, outcome window, label registry, maturity. Recall task: Draft the minimum contract for a five-session equity return target and mark every choice that affects economic meaning.

Decision time, entry time, and label start

INTERMEDIATE

ONE-SENTENCE MEANING

Decision time is when the model commits to an action, entry time is the modeled executable fill, and label start is the price reference used to measure the outcome. These times may differ because computation, order routing, auctions, and market latency consume time.

MEMORY JOGGER

The label should begin where the strategy could actually own risk, not where the feature was observed. Model the delay paid by the intended workflow and keep it identical in research and simulation.

WHY IT MATTERS IN QUANT RESEARCH

Using the same close for features, decision, and entry often assumes an impossible fill after observing the final print. A few minutes of hidden latency can explain much of a short-horizon backtest.

FORMULA INTUITION

$t_{\text{feature}} \leq t_{\text{decision}} < t_{\text{entry}} \leq t_{\text{exit}}$, with $y_t = P_{\text{exit}}/P_{\text{entry}} - 1$ under a declared fill rule. If an auction is used, its cutoff and eligibility rules become part of the target.

SIMPLE MARKET EXAMPLE

A signal computed from the official 16:00 close cannot normally buy that exact close. A next-open target is causal but includes overnight risk; a close-auction target requires features frozen before the auction order deadline.

PYTHON / SYSTEM IMPLEMENTATION

Represent all clocks explicitly and create an execution calendar that maps each decision to eligible entry and exit events. Test daylight-saving changes, half days, halts, and missing auctions rather than assuming a fixed row lag.

CONFUSIONS TO AVOID

Lagging the label by one record does not repair an impossible same-bar fill when records have different publication times. A timestamp at 16:00 also does not prove the final bar was available at 16:00 exactly.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scheduler defines prediction cutoffs, the execution simulator maps them to fills, and the label builder consumes those fills. This separation lets latency assumptions change without silently rewriting feature causality.

RELATED CONCEPTS AND QUICK RECALL

Related: availability time, execution lag, auction cutoff, fill model. Recall task: Explain why a close-derived signal needs a later entry reference and identify the risk introduced by using the next open.

Forecast horizon versus holding period

INTERMEDIATE

ONE-SENTENCE MEANING

Forecast horizon is the future interval summarized by the target, while holding period is how long a realized position remains open. They coincide only for simple one-shot strategies without refresh, scaling, overlap, or exit logic.

MEMORY JOGGER

The model predicts an outcome window; the portfolio implements a sequence of trades. Distinguish what is estimated from how forecasts are translated into inventory.

WHY IT MATTERS IN QUANT RESEARCH

Confusing the two can understate turnover, double-count overlapping exposure, or evaluate a model on an outcome it never monetizes. Horizon choice should reflect signal decay, execution latency, costs, and rebalance cadence.

FORMULA INTUITION

A q -session target is $y_t = R_{(t+1:t+q)}$, but daily forecasts can create q overlapping vintages. Portfolio exposure at date u may combine forecasts issued at $u, u-1, \dots, u-q+1$.

SIMPLE MARKET EXAMPLE

A model predicts ten-day residual return every day, yet the portfolio exponentially blends daily scores and holds names for roughly six days. Ten days describes the label; six days describes inventory persistence.

PYTHON / SYSTEM IMPLEMENTATION

Record target horizon, prediction frequency, score half-life, rebalance policy, and realized holding distribution separately. Backtests should attribute PnL by forecast vintage to show whether the intended horizon is actually captured.

CONFUSIONS TO AVOID

Dividing annual turnover by a nominal horizon does not recover the realized holding period. A target can be long horizon while stop losses, risk limits, or new forecasts cause much earlier exits.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Research estimates decay across horizons, portfolio construction chooses score blending, and execution records actual open-to-close durations. Monitoring compares realized capture with the target's payoff curve.

RELATED CONCEPTS AND QUICK RECALL

Related: signal decay, rebalance frequency, forecast vintage, turnover. Recall task: Give an example in which a ten-day target leads to a shorter realized holding period and explain why that is not inconsistent.

Point, interval, and path-dependent outcomes

INTERMEDIATE

ONE-SENTENCE MEANING

A point target uses selected endpoints, an interval target aggregates events across a window, and a path-dependent target depends on the route taken between entry and exit. The choice determines which risks and trading behaviors the model can learn.

MEMORY JOGGER

Endpoints answer where the market finished; paths answer how it got there. Use the least complex target that still represents the decision and risk rule.

WHY IT MATTERS IN QUANT RESEARCH

Two trades with identical final return can have very different drawdown, opportunity, and stop behavior. Endpoint labels are efficient for broad forecasting, while path labels are necessary for barriers, excursions, and execution-aware exits.

FORMULA INTUITION

Point: P_e/P_{s-1} . Interval: sum or compound eligible returns over $(s,e]$. Path: $L(\{P_u: s \leq u \leq e\})$, such as first barrier hit or maximum adverse excursion.

SIMPLE MARKET EXAMPLE

One stock rises 4%, falls 9%, then closes up 2%; another moves smoothly to the same 2% gain. A fixed-horizon return treats them equally, but a stop-based target labels the first path as a loss.

PYTHON / SYSTEM IMPLEMENTATION

Build path targets from ordered event sequences with explicit sampling, high-low semantics, and tie-breaking. Preserve entry, exit, extrema, first-hit time, and reason code so the label can be audited.

CONFUSIONS TO AVOID

Daily high and low do not reveal which barrier was hit first when both appear in one bar. Increasing path detail can also make labels more sensitive to bad ticks and microstructure noise.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label service offers endpoint and path-aware families, while strategy validation chooses the version matching exit mechanics. Risk attribution uses excursion fields even when the predictive target remains endpoint-based.

RELATED CONCEPTS AND QUICK RECALL

Related: maximum excursion, first passage, barrier label, endpoint return. Recall task: Describe two price paths with the same endpoint return but opposite stop-loss outcomes.

Economic target versus statistical proxy

INTERMEDIATE

ONE-SENTENCE MEANING

An economic target is the outcome the strategy ultimately values, while a statistical proxy is a more observable or learnable quantity believed to help produce that outcome. A proxy is acceptable only when its link to the decision is tested and monitored.

MEMORY JOGGER

Predict what can be learned, but optimize what matters. Keep the bridge from proxy accuracy to net trading value explicit rather than assuming every predictive gain is monetizable.

WHY IT MATTERS IN QUANT RESEARCH

Mid-price direction, quote movement, realized volatility, and fill probability may be easier to model than net PnL. Their usefulness depends on how a policy converts them into orders, exposure, and risk-adjusted return.

FORMULA INTUITION

Economic utility $U = E[\text{PnL} - \text{costs} - \text{risk penalties}]$. A proxy z is useful when decisions based on $E[z|x]$ improve out-of-sample U under the implemented policy.

SIMPLE MARKET EXAMPLE

A market-making model predicts the next mid-price tick with 58% accuracy, but aggressive fills erase the edge. A second model with lower direction accuracy may earn more by predicting adverse selection conditional on fill.

PYTHON / SYSTEM IMPLEMENTATION

Version proxy definitions beside the downstream policy, then run policy-level evaluation and mediation diagnostics. Track whether changes in proxy metrics produce expected changes in fills, costs, exposure, and net return.

CONFUSIONS TO AVOID

High R-squared on a proxy does not prove economic value, and directly using net PnL as a label can inject policy-specific noise. A proxy can also become invalid when market mechanics or execution behavior changes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Model development may optimize the proxy, but promotion gates use strategy-level outcomes. The monitoring layer watches both proxy calibration and the proxy-to-utility relationship.

RELATED CONCEPTS AND QUICK RECALL

Related: surrogate objective, utility, policy evaluation, adverse selection. Recall task: Name a useful market microstructure proxy and state the evidence needed to show it improves net economic utility.

Target units, scaling, and invariance

INTERMEDIATE

ONE-SENTENCE MEANING

Target units determine what errors mean and how observations compare across assets, time, and regimes. Scaling should create the invariance required by the hypothesis without erasing economically meaningful magnitude.

MEMORY JOGGER

Ask whether the model should predict dollars, percent, basis points, risk units, ranks, or probabilities. Every loss function inherits the target's unit and therefore its implicit weighting.

WHY IT MATTERS IN QUANT RESEARCH

Raw dollar moves overweight high-priced assets, while unscaled returns can let volatile instruments dominate squared error. Risk scaling improves comparability but changes the objective toward standardized rather than absolute opportunity.

FORMULA INTUITION

Return $R = \Delta P/P$; basis points = $10,000R$; risk-scaled $y = R/\sigma_{\text{prior}}$; dollar PnL = quantity $\times$ multiplier $\times$ ΔP . The scaling denominator must be known at decision time.

SIMPLE MARKET EXAMPLE

A 1% move in a 10% annualized-volatility bond ETF and a 1% move in an 80% volatility biotech stock are different standardized outcomes. The raw returns are equal, but their surprise and sizing implications differ.

PYTHON / SYSTEM IMPLEMENTATION

Attach unit metadata, price convention, currency, multiplier, and scale artifact to every target. Fit denominators on prior data, apply floors, and expose both raw and scaled outcomes for interpretation.

CONFUSIONS TO AVOID

Standardizing with future or full-sample volatility leaks regime information. Converting everything to ranks removes magnitude and can reward correct ordering even when no spread survives trading costs.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label layer publishes compatible units to model losses, metrics, portfolio maps, and attribution. Cross-asset systems use explicit conversion stages so economic PnL remains reconstructable.

RELATED CONCEPTS AND QUICK RECALL

Related: dimensionless return, volatility scaling, basis point, loss weighting. Recall task: Choose units for a cross-asset forecasting problem and explain what observations the chosen loss will emphasize.

Outcome eligibility, censoring, and missing labels

INTERMEDIATE

ONE-SENTENCE MEANING

A label is missing when its outcome cannot be observed under the contract, while censoring means only partial information about the outcome is known. Treating either state as an ordinary zero changes the estimand and often selects an unrealistic survivor population.

MEMORY JOGGER

Missing is not flat, and unavailable is not a loss. Record why the outcome is incomplete and whether the observation remains usable under a specialized likelihood or policy.

WHY IT MATTERS IN QUANT RESEARCH

Delistings, halts, stale quotes, contract expiry, corporate actions, and incomplete future windows all create label gaps. Dropping them without analysis can remove exactly the stressed cases that drive strategy risk.

FORMULA INTUITION

$\text{eligibility}_{i,t} = 1$ only if entry and required outcome observations exist under policy. For right-censored time T , observe $\min(T, C)$ plus indicator $\delta = 1[T \leq C]$.

SIMPLE MARKET EXAMPLE

A stock disappears before the planned exit because of an acquisition. Deleting the row biases toward continuing names; applying a documented cash-merger settlement may recover an economic return, while an unresolved halt remains censored.

PYTHON / SYSTEM IMPLEMENTATION

Emit value, eligibility, maturity, censor flag, reason code, last observed time, and recovery method. Report exclusion rates by cohort and outcome, then use inverse-probability or survival methods only when their assumptions are defensible.

CONFUSIONS TO AVOID

Forward-filling an exit price fabricates an executable outcome. Complete-case analysis is not neutral when disappearance depends on distress, illiquidity, or the target itself.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Reference-data and corporate-action services resolve known exits; the label builder classifies unresolved cases. Validation dashboards show coverage and selection effects before model metrics are accepted.

RELATED CONCEPTS AND QUICK RECALL

Related: right censoring, survivor bias, delisting return, eligibility. Recall task: Explain why setting an unresolved future return to zero is wrong and propose an auditable alternative state.

Label maturity and revision vintages

INTERMEDIATE

ONE-SENTENCE MEANING

A label matures only after its full outcome window and required source corrections are available. Later corporate-action, benchmark, or trade corrections may create a new vintage without changing what was known when the prediction was made.

MEMORY JOGGER

Prediction-time data must be as-known; evaluation-time outcomes may be provisional or final. Keep those two vintage policies separate and never rewrite past experiment identity silently.

WHY IT MATTERS IN QUANT RESEARCH

Long horizons delay feedback, and premature evaluation overweights easy-to-mature rows near the start of a batch. Revised outcomes can also change rankings, class thresholds, and model-comparison conclusions.

FORMULA INTUITION

$\text{mature}(t_{\text{eval}}) = 1$ when $\text{exit_time} + \text{settlement_or_revision_lag} \leq t_{\text{eval}}$. A bitemporal label has an economic interval plus a knowledge interval for each outcome vintage.

SIMPLE MARKET EXAMPLE

A twenty-session target issued on June 1 is not eligible for final scoring on June 10. A split adjustment posted later may correct both entry and exit prices, producing a new label artifact while preserving the original experiment snapshot.

PYTHON / SYSTEM IMPLEMENTATION

Materialize provisional and final states with vintage IDs, maturity timestamps, revision reasons, and supersession links. Freeze experiment datasets or retain exact source snapshots so reported results can be replayed.

CONFUSIONS TO AVOID

Using whatever labels are currently mature creates a changing evaluation sample. Overwriting a target table after a vendor correction destroys the evidence behind earlier research decisions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evaluation scheduler scores only mature cohorts, the registry tracks revisions, and model governance specifies whether thresholds respond to corrected history. Live dashboards separate forecast quality awaiting outcomes from completed performance.

RELATED CONCEPTS AND QUICK RECALL

Related: bitemporal data, provisional outcome, revision policy, dataset snapshot. Recall task: State when a twenty-session label becomes scoreable and how a later data correction should be represented.

Simple, log, and compounded forward return

INTERMEDIATE

ONE-SENTENCE MEANING

Simple return measures wealth change, log return measures the price-ratio logarithm, and compounded return combines subperiod returns multiplicatively; differences grow with large moves.

MEMORY JOGGER

Use simple return for investor wealth and log return for additive time-series algebra. Declare total-return treatment, currency, endpoints, and whether interim cash flows are reinvested.

WHY IT MATTERS IN QUANT RESEARCH

Return definition affects loss scale, tail symmetry, and multi-period aggregation. Mixing summed simple returns with compounded realized PnL creates incorrect labels precisely in volatile periods.

FORMULA INTUITION

$R_{(s,e)} = P_e/P_s - 1$; $r_{(s,e)} = \log(P_e/P_s)$; $1 + R_{(s,e)} = \text{product}_u(1 + R_u)$. Log returns add: $r_{(s,e)} = \sum_u r_u$.

SIMPLE MARKET EXAMPLE

A rise from 100 to 110 followed by a fall to 99 gives +10% then -10%. The sum is zero, but compounded wealth is 0.99 and the total simple return is -1%; log returns sum to $\log(0.99)$.

PYTHON / SYSTEM IMPLEMENTATION

Build returns from an approved total-return series or explicit cash-flow logic and verify endpoint identity. Round-trip tests should compare direct price ratios, compounded subperiod returns, and summed logs within numerical tolerance.

CONFUSIONS TO AVOID

Arithmetic and log returns are close only for small moves. Log return is undefined for nonpositive prices and should not be used to hide data defects or instruments with different payoff structures.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The canonical return service publishes reusable outcome primitives consumed by labels, risk, and attribution. Target versions reference the primitive rather than reimplementing price arithmetic in each notebook.

RELATED CONCEPTS AND QUICK RECALL

Related: total return, gross return, cash flow, log additivity. Recall task: Calculate the two-period result of +10% and -10% and explain why adding simple returns fails.

Direction and sign classification

INTERMEDIATE

ONE-SENTENCE MEANING

A direction label maps a continuous outcome to up or down, often with a policy for exact zeros. It simplifies the prediction task but discards magnitude, making accuracy an incomplete measure of trading value.

MEMORY JOGGER

A correct tiny move and a correct crash count equally unless weighting or utility restores magnitude. Use sign labels only when the downstream action truly depends mainly on direction.

WHY IT MATTERS IN QUANT RESEARCH

Direction models can be robust when magnitude is noisy, and predicted probabilities support calibrated position rules. Yet class prevalence, transaction costs, and asymmetric losses determine whether more than 50% accuracy matters.

FORMULA INTUITION

$y=1[R_{\text{net}}>0]$ for binary labels, with $P(y=1|x)$ as the forecast. A cost-aware rule trades only when expected payoff $p*g-(1-p)*l$ -cost exceeds zero.

SIMPLE MARKET EXAMPLE

A model correctly predicts nine gains of 1 basis point and misses one loss of 30 basis points. Its 90% accuracy is economically poor, demonstrating why sign and payoff magnitude must be evaluated together.

PYTHON / SYSTEM IMPLEMENTATION

Specify zero handling, net or gross return, class encoding, and prevalence by period. Report calibration, balanced metrics, payoff-weighted results, and simulated policy value rather than accuracy alone.

CONFUSIONS TO AVOID

Relabeling zero as down can create artificial imbalance in illiquid assets with repeated prices. A probability threshold of 0.5 is not automatically optimal when gains, losses, and costs are asymmetric.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The classifier produces probabilities, the decision policy applies utility and constraints, and attribution decomposes hit rate from average gain and loss. Monitoring watches prevalence and calibration drift.

RELATED CONCEPTS AND QUICK RECALL

Related: binary classification, probability calibration, hit rate, payoff asymmetry. Recall task: Construct a high-accuracy but money-losing sign forecast and identify the missing economic statistic.

Threshold and three-way labels

INTERMEDIATE

ONE-SENTENCE MEANING

Threshold labels separate materially positive, neutral, and materially negative outcomes using bands tied to noise, spread, risk, or economic relevance. The neutral class prevents tiny indistinguishable moves from dominating directional training.

MEMORY JOGGER

Do not force every observation into a trade. Set the dead zone from a causal decision rule and evaluate how sensitive results are to its size.

WHY IT MATTERS IN QUANT RESEARCH

Three-way labels can focus capacity on actionable moves and reduce label noise near zero. Threshold choice also changes prevalence and may become a hidden hyperparameter selected for flattering backtests.

FORMULA INTUITION

$y=+1$ if $R > u_t$, $y=0$ if $l_t \leq R \leq u_t$, and $y=-1$ if $R < l_t$. Symmetric risk-scaled bands often use $u_t = k \cdot \sigma_{\text{prior}}$ and $l_t = -k \cdot \sigma_{\text{prior}}$.

SIMPLE MARKET EXAMPLE

With expected round-trip cost of 8 basis points, returns inside ± 8 basis points may be economically neutral. A 20 basis point forecast then counts as positive, while a 3 basis point move does not reward a forced long call.

PYTHON / SYSTEM IMPLEMENTATION

Fit threshold parameters inside training folds, log resulting class shares, and keep raw continuous outcomes for secondary analysis. Compare fixed, cost-based, and volatility-based bands across regimes and liquidity cohorts.

CONFUSIONS TO AVOID

Using realized future volatility to set the threshold leaks the outcome window. Optimizing the band on final test performance converts the label itself into an unreported model search.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Label configuration and cost assumptions share a versioned dependency. The strategy layer may abstain or reduce size when predicted neutral probability is high.

RELATED CONCEPTS AND QUICK RECALL

Related: dead zone, abstention, class prevalence, cost hurdle. Recall task: Design a three-way label for a strategy with an 8 basis point cost hurdle and state which inputs must be causal.

Volatility-scaled return targets

INTERMEDIATE

ONE-SENTENCE MEANING

A volatility-scaled target divides future return by a risk estimate known before entry, expressing outcomes in ex-ante risk units. It makes errors more comparable across assets but shifts the estimand away from raw return magnitude.

MEMORY JOGGER

Scale by risk available at the decision, not risk realized inside the label window. Use floors and stale-state rules because a tiny denominator can create extreme labels.

WHY IT MATTERS IN QUANT RESEARCH

Cross-sectional models otherwise devote large loss weight to naturally volatile names. Scaling can improve stability and align with risk-balanced portfolios, while still requiring raw-PnL evaluation.

FORMULA INTUITION

$y_t = R_{(t+1:t+q)} / \max(\sigma_{\hat{t}} \sqrt{q}, \sigma_{\text{floor}})$. This assumes square-root horizon scaling only as an approximation and does not remove serial dependence.

SIMPLE MARKET EXAMPLE

A 2% five-day return with forecast five-day volatility of 1% equals +2 risk units; the same return with 4% risk equals +0.5. The first outcome is more unusual under the prior state.

PYTHON / SYSTEM IMPLEMENTATION

Version the volatility estimator, lookback, annualization, lag, and floor with the label. Store numerator and denominator separately and audit target tails by denominator bucket.

CONFUSIONS TO AVOID

Dividing by realized future volatility uses information from the outcome and can reward low-volatility paths mechanically. Risk scaling also does not guarantee equal economic capacity or equal trading costs.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk service supplies point-in-time forecasts, the label builder creates standardized outcomes, and portfolio evaluation converts predictions back to expected returns and positions. Monitoring compares scaled and raw performance.

RELATED CONCEPTS AND QUICK RECALL

Related: ex-ante volatility, sigma floor, standardized return, heteroskedasticity. Recall task: Explain why future realized volatility is an invalid denominator and calculate two risk-scaled outcomes from equal raw returns.

Excess and benchmark-relative targets

INTERMEDIATE

ONE-SENTENCE MEANING

An excess target subtracts a financing rate, market return, sector return, or custom benchmark from the asset outcome. It directs the model toward relative performance rather than broad beta that may be easier to predict but unwanted.

MEMORY JOGGER

Name the benchmark and its tradable timing. Relative return is meaningful only when both legs use aligned calendars, currencies, horizons, and executable references.

WHY IT MATTERS IN QUANT RESEARCH

Market-neutral and factor strategies care about idiosyncratic return, while asset allocators may intentionally want systematic exposure. Benchmark choice therefore encodes the investment mandate, not just a statistical cleanup.

FORMULA INTUITION

$y_i = R_i - R_b$ for arithmetic excess return, or $y_i = R_i - \beta_{\hat{i}} R_m$ for beta-adjusted return. The beta estimate must use prior data and the chosen convention is not algebraically identical across compounding methods.

SIMPLE MARKET EXAMPLE

A stock gains 3% while its sector gains 5%. Its raw label is positive, but sector-relative performance is -2%, which is the relevant outcome for a sector-neutral long-short portfolio.

PYTHON / SYSTEM IMPLEMENTATION

Use synchronized total-return series and version benchmark membership point in time. Publish raw, benchmark, and excess components so changes can be traced to the asset or reference leg.

CONFUSIONS TO AVOID

Subtracting today's final sector return from an intraday target can introduce asynchronous information. A residual target estimated with a full-sample beta leaks future exposure and may over-neutralize genuine signal.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Benchmark services provide investable reference outcomes; model training consumes the relative target; portfolio attribution verifies that predicted alpha remains after realized factor exposures and costs.

RELATED CONCEPTS AND QUICK RECALL

Related: active return, beta adjustment, sector neutralization, financing rate. Recall task: Reinterpret a +3% stock return when its sector gained 5% and state when raw return would still be the preferred target.

Cross-sectional rank and quantile targets

INTERMEDIATE

ONE-SENTENCE MEANING

A rank target encodes an asset's relative position within an eligible contemporaneous universe, while quantile labels group those ranks into buckets. These targets emphasize ordering and reduce sensitivity to market-wide shocks and extreme magnitudes.

MEMORY JOGGER

Rank only assets that were simultaneously eligible under the strategy's universe. The peer set, tie rule, minimum breadth, and missing policy are part of every label value.

WHY IT MATTERS IN QUANT RESEARCH

Cross-sectional portfolios often monetize spreads between winners and losers rather than absolute forecasts. Rank targets align with that construction, but can hide whether the entire universe lost money or the spread is too small to trade.

FORMULA INTUITION

$\text{rank_pct}_i = (\text{rank}_i - 1) / (N - 1)$, and $\text{quantile}_i = Q(\text{rank_pct}_i)$. A demeaned alternative $y_i = R_i - \text{mean}(j \text{ in } U_t)R_j$ preserves some magnitude.

SIMPLE MARKET EXAMPLE

On a broad down day, a stock returning -1% may rank in the top decile when peers lose 4%. The rank correctly signals relative strength but does not imply a profitable unhedged long position.

PYTHON / SYSTEM IMPLEMENTATION

Snapshot point-in-time universe membership, compute outcomes first, then rank within valid groups after the horizon matures. Store group ID, breadth, tie method, raw outcome, and percentile.

CONFUSIONS TO AVOID

Using today's index constituents for historical labels creates survivor bias. Quantile buckets can also jump when universe breadth changes or many illiquid names share zero returns.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The universe service defines peers, the label builder produces rank and raw components, and portfolio evaluation tests long-short spreads plus hedge requirements. Monitoring tracks group breadth and bucket stability.

RELATED CONCEPTS AND QUICK RECALL

Related: percentile rank, peer group, quantile portfolio, cross-sectional spread. Recall task: Explain how a negative raw return can receive a top-decile label and why that label does not justify an unhedged long.

Residual and factor-neutral targets

INTERMEDIATE

ONE-SENTENCE MEANING

A residual target removes returns explained by selected factors using exposures estimated from information available before the outcome. It asks the model to forecast idiosyncratic performance conditional on a stated risk model.

MEMORY JOGGER

Residual means unexplained by the chosen model, not free of risk. The factor set, exposure estimator, estimation window, and rebalance lag define the target.

WHY IT MATTERS IN QUANT RESEARCH

Neutral labels can prevent a stock model from rediscovering market, sector, duration, or style bets. They may also remove economically useful structure or become unstable when exposures shift during stressed markets.

FORMULA INTUITION

$y_{i,t} = R_{i,t} - \beta_{\hat{i},t}' F_t$, where $\beta_{\hat{i},t}$ is fixed before the label window and F_t is the realized factor return over that window. Cross-sectional residualization is another distinct convention.

SIMPLE MARKET EXAMPLE

A bank stock gains 4% while the banking factor gains 3.5%; a prior beta of 1.1 implies factor contribution of 3.85%, leaving about 0.15% residual. The label isolates stock-specific movement under that model.

PYTHON / SYSTEM IMPLEMENTATION

Version factor definitions, exposure snapshots, constraints, and residualization method. Include raw return, predicted factor component, residual, and exposure age so the target can be reconstructed.

CONFUSIONS TO AVOID

Estimating beta with returns that include the outcome window leaks future co-movement. Treating residuals from different risk-model versions as one target also invalidates longitudinal comparisons.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk model supplies point-in-time exposures; labels combine them with future factor outcomes; portfolio attribution tests whether predicted residuals remain neutral after implementation.

RELATED CONCEPTS AND QUICK RECALL

Related: risk model, factor exposure, idiosyncratic return, neutralization. Recall task: Compute a simple factor residual and identify which part of the calculation must be frozen before the forecast.

Maximum favorable and adverse excursion

INTERMEDIATE

ONE-SENTENCE MEANING

Maximum favorable excursion records the best mark-to-market gain within a window, while maximum adverse excursion records the worst loss. Together they summarize path opportunity and pain without specifying a single exit rule.

MEMORY JOGGER

MFE asks how far the trade could have worked; MAE asks how far it moved against the position. Both depend on side, sampling frequency, executable price convention, and bad-tick controls.

WHY IT MATTERS IN QUANT RESEARCH

Excursions help design stops, profit targets, and sizing rules, and reveal whether endpoint losses contained temporary opportunity. They are optimistic if computed from unattainable highs or lows without execution constraints.

FORMULA INTUITION

For a long, $MFE = \max_u (P_u / P_{\text{entry}} - 1)$ and $MAE = \min_u (P_u / P_{\text{entry}} - 1)$ over the declared path. For a short, signs and executable bid-ask references must be defined consistently.

SIMPLE MARKET EXAMPLE

A long enters at 100, trades to 104, falls to 96, and exits at 101. Endpoint return is +1%, MFE is +4%, and MAE is -4%, exposing a path that the final return hides.

PYTHON / SYSTEM IMPLEMENTATION

Generate excursions from cleaned ordered events, record time-to-extreme, and decide whether bars use high-low or bid-ask executable bounds. Validate against tick-level samples and flag simultaneous extreme ambiguity.

CONFUSIONS TO AVOID

Daily highs and lows may not be executable at size and do not reveal ordering. Choosing stop parameters after inspecting MFE and MAE distributions can overfit the same paths later used for evaluation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Research uses excursion labels to design exit hypotheses; simulation applies actual rules; risk uses MAE distributions for stress and capital limits. The raw path remains preserved for replay.

RELATED CONCEPTS AND QUICK RECALL

Related: run-up, drawdown, stop loss, path dependence. Recall task: Calculate MFE and MAE for a long that moves from 100 to 104 to 96 and exits at 101.

Barrier and triple-barrier labels

INTERMEDIATE

ONE-SENTENCE MEANING

Barrier labels classify an outcome by which price or risk boundary is reached first before a maximum time limit. The triple-barrier method uses an upper profit barrier, lower loss barrier, and vertical time barrier to encode path-dependent exits.

MEMORY JOGGER

The first hit determines the event, not the best endpoint. Barriers should reflect causal volatility, costs, side, and executable prices rather than a tuned chart pattern.

WHY IT MATTERS IN QUANT RESEARCH

This label family aligns training with stop-profit-time exit logic and adapts thresholds to asset risk. It is sensitive to sampling frequency, within-bar order, and the method used to assign concurrent hits.

FORMULA INTUITION

$\text{upper} = P_{\text{entry}}(1 + k_u \sigma_{\text{prior}})$, $\text{lower} = P_{\text{entry}}(1 - k_l \sigma_{\text{prior}})$, $\text{vertical} = t + q$. Label is +1, -1, or terminal outcome according to the earliest valid crossing.

SIMPLE MARKET EXAMPLE

With entry 100 and barriers at 103 and 97, a path touching 97 on day two then ending at 105 receives the lower-barrier label. The final gain does not erase the earlier stop event.

PYTHON / SYSTEM IMPLEMENTATION

Create an event scanner with ordered timestamps, side-aware bid-ask prices, gap rules, tie-breaking, and causal volatility inputs. Store first-hit time, barrier values, crossing price, terminal return, and ambiguity code.

CONFUSIONS TO AVOID

A daily bar showing high 104 and low 96 cannot reveal which barrier hit first. Optimizing barrier multipliers on the evaluation set is label selection bias, not just model tuning.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event-label service converts a declared trading hypothesis into outcomes; validation uses purge intervals spanning each event; execution simulation checks whether barrier fills are realistic after gaps and costs.

RELATED CONCEPTS AND QUICK RECALL

Related: first passage, vertical barrier, event sampling, stop-profit rule. Recall task: Label a path that hits the lower barrier before later crossing the upper barrier and explain the within-bar ambiguity problem.

Event-driven and opportunity labels

INTERMEDIATE

ONE-SENTENCE MEANING

Event-driven labels create training examples only when a predeclared trigger makes a decision relevant, such as an earnings release, volatility shock, spread divergence, or order-book imbalance. They change the sampled population as well as the outcome definition.

MEMORY JOGGER

First define when the system is allowed to ask a question. Then measure the future outcome from a causal entry tied to that event.

WHY IT MATTERS IN QUANT RESEARCH

Sampling every timestamp can flood training with redundant quiet observations and obscure conditional opportunities. Event sampling can improve focus, but a trigger tuned on outcomes leaks selection and limits generalization.

FORMULA INTUITION

$\text{sample_t}=1$ when $\text{trigger}(x_{\leq t})=1$ and cooldown rules permit an event. Target y_t then uses an outcome window beginning after the trigger's availability and modeled execution delay.

SIMPLE MARKET EXAMPLE

A post-earnings model creates one example after a filing becomes machine-readable, rather than one label for every preceding minute. Its horizon may begin at the next tradable session after parsing completes.

PYTHON / SYSTEM IMPLEMENTATION

Version trigger logic, thresholds, cooldown, deduplication, event family, and source clocks. Retain non-events for prevalence diagnostics and build group-aware splits when one event generates multiple related observations.

CONFUSIONS TO AVOID

Defining events with future peak volatility or realized price moves conditions on the answer. Multiple triggers around the same shock are not independent samples and should not be scattered across train and test.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event detector creates immutable opportunity IDs; feature and label builders attach to those IDs; validation groups related events; deployment monitors trigger rate and coverage drift.

RELATED CONCEPTS AND QUICK RECALL

Related: event sampling, trigger, cooldown, opportunity set. Recall task: Design a causal earnings event and identify the earliest executable label start after the source becomes machine-readable.

Realized volatility and risk targets

INTERMEDIATE

ONE-SENTENCE MEANING

A realized-volatility target summarizes future return variation rather than direction, using squared returns, absolute returns, ranges, or higher-frequency estimators. Its definition must handle market hours, jumps, microstructure noise, and annualization consistently.

MEMORY JOGGER

Predict the amount of movement, not its sign. The sampling interval and estimator determine which variation is treated as signal versus noise.

WHY IT MATTERS IN QUANT RESEARCH

Risk forecasts drive position sizing, option valuation, limits, and regime controls. A target contaminated by bad ticks or inconsistent overnight treatment trains the model to predict data mechanics rather than risk.

FORMULA INTUITION

$RV_{(t,t+q)} = \sqrt{\sum u^2}$ over eligible subperiods, optionally annualized. Range and kernel estimators have different efficiency and noise assumptions, so they require separate target IDs.

SIMPLE MARKET EXAMPLE

Four future intraday returns of 1%, -1%, 0.5%, and -0.5% yield sum of squares 0.00025 and realized volatility $\sqrt{0.00025} = 1.581\%$ for that window. Direction cancels in return but not in variance.

PYTHON / SYSTEM IMPLEMENTATION

Use a canonical calendar, cleaned high-frequency returns, overnight policy, and minimum observation count. Publish estimator components, sample count, jump flags, and annualization basis with the target.

CONFUSIONS TO AVOID

Taking the square root before summing is not realized volatility, and close-to-close squared return is not equivalent to intraday variation. More frequent sampling can increase microstructure noise rather than improve accuracy.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk label service supports volatility models, while portfolio systems compare predicted risk with subsequent realized risk and PnL scaling. Monitoring separates level error, calibration, and tail underprediction.

RELATED CONCEPTS AND QUICK RECALL

Related: realized variance, range estimator, jump variation, annualization. Recall task: Compute realized volatility from four subperiod returns and explain why their signed sum is irrelevant to the risk target.

Multi-horizon and multi-task targets

INTERMEDIATE

ONE-SENTENCE MEANING

Multi-horizon targets measure one outcome over several windows, while multi-task targets jointly represent related outcomes such as return, volatility, direction, and drawdown.

MEMORY JOGGER

Treat horizons as a curve, not independent contests. Shared models should learn common structure while allowing each task to retain its own unit, loss, and uncertainty.

WHY IT MATTERS IN QUANT RESEARCH

A signal may forecast one-day reversal and twenty-day continuation with opposite signs. Joint targets can reveal this shape and support portfolio timing, but headline metrics must not average away task failures.

FORMULA INTUITION

$Y_t = [R_{(t+1:t+1)}, R_{(t+1:t+5)}, R_{(t+1:t+20)}]$ or tasks $[R, RV, \text{hit}]$. A combined loss $\sum_k \lambda_k L_k$ requires weights that define relative importance.

SIMPLE MARKET EXAMPLE

An earnings surprise model may predict strong five-day residual return, flat twenty-day return, and elevated ten-day volatility. Those outputs support different entry, holding, and sizing decisions.

PYTHON / SYSTEM IMPLEMENTATION

Store one base decision ID with separate outcome windows, maturity states, masks, units, and loss weights. Validate each head, cross-task consistency, and the downstream policy rather than only the aggregate training loss.

CONFUSIONS TO AVOID

Longer horizons mature later and can create a changing task mix near dataset boundaries. Selecting the best horizon after viewing the test set is multiple testing even when all heads share one model.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The target graph publishes related outcomes; training handles missing maturity masks; portfolio logic consumes horizon-specific forecasts; governance records which outputs are decision-critical.

RELATED CONCEPTS AND QUICK RECALL

Related: term structure, auxiliary task, loss weight, shared representation. Recall task: Sketch a three-horizon return target and explain why choosing the best test horizon is a selection problem.

Time-to-event and survival targets

INTERMEDIATE

ONE-SENTENCE MEANING

A time-to-event target estimates when a market or operational event occurs while correctly handling observations whose event has not happened by the study end. It separates event probability from timing and preserves censored cases.

MEMORY JOGGER

Observe duration plus whether the event was seen. Do not convert every unfinished case into a permanent non-event.

WHY IT MATTERS IN QUANT RESEARCH

Applications include time to barrier hit, default, spread normalization, order fill, or volatility breakout. Survival methods use incomplete paths efficiently when censoring assumptions and competing events are stated.

FORMULA INTUITION

Observe $T_{\text{obs}} = \min(T_{\text{event}}, T_{\text{censor}})$ and $\text{delta} = 1[T_{\text{event}} \leq T_{\text{censor}}]$. Hazard $h(t|x)$ describes instantaneous event intensity conditional on survival to t .

SIMPLE MARKET EXAMPLE

An order remains unfilled when the five-minute observation window ends. It is right-censored at five minutes, not labeled as never fill; a cancellation may be a competing event rather than ordinary censoring.

PYTHON / SYSTEM IMPLEMENTATION

Store start, observed duration, event type, censor type, last observation, and eligibility. Use group-aware splits, time-dependent metrics, and competing-risk methods when one event prevents another.

CONFUSIONS TO AVOID

Binary 'filled within five minutes' labels discard information about a fill at ten seconds versus four minutes. Treating cancellation as independent censoring can be wrong when traders cancel because fill probability deteriorates.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event engine records state transitions; survival labels feed timing models; execution policy translates hazards into order updates; monitoring compares predicted and realized event curves.

RELATED CONCEPTS AND QUICK RECALL

Related: hazard, right censoring, competing risk, duration. Recall task: Explain how to encode an unfilled order at the end of a five-minute window and how cancellation complicates censoring.

Overlapping forward returns

INTERMEDIATE

ONE-SENTENCE MEANING

Forward labels overlap when adjacent examples share part of the same future outcome window. This creates serial dependence in targets and allows the same realized return to influence multiple training or evaluation observations.

MEMORY JOGGER

A five-day label issued daily contains four days in common with tomorrow's label. More rows do not mean proportionally more independent evidence.

WHY IT MATTERS IN QUANT RESEARCH

Overlap inflates naive significance, smooths metric series, and creates leakage when train and test windows share outcome events. It is a property of the labeling geometry, not a defect fixed by shuffling.

FORMULA INTUITION

$y_t = \sum_{u=t+1}^{t+q} r_u$, so y_t and y_{t+1} share $q-1$ subperiod returns. Under independent r , $\text{Corr}(y_t, y_{t+1})$ is approximately $(q-1)/q$.

SIMPLE MARKET EXAMPLE

For $q=5$, adjacent labels share four of five daily returns and have approximate correlation 0.8 under simple assumptions. A single crash day appears inside five consecutive labels.

PYTHON / SYSTEM IMPLEMENTATION

Record each label's start and end interval, use interval-aware splitters, and estimate uncertainty with blocks or event clusters. Visualize how many labels include each outcome date to detect concentration.

CONFUSIONS TO AVOID

Random train-test splits are especially invalid because near-identical overlapping outcomes land on both sides. Thinning to every q -th sample reduces overlap but can waste information and create arbitrary phase dependence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label table exposes intervals to the validation engine, which purges conflicting examples and uses robust inference. Reporting distinguishes raw row count from effective evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: serial dependence, effective sample size, interval overlap, block bootstrap. Recall task: For a five-day target issued daily, count shared days between adjacent labels and explain why row count overstates evidence.

Purging label intervals

INTERMEDIATE

ONE-SENTENCE MEANING

Purging removes training examples whose outcome intervals overlap the validation or test period. It prevents realized events used in evaluation from also appearing inside training labels.

MEMORY JOGGER

Split on information intervals, not only feature timestamps. Any training target that reaches into the held-out outcome region must be excluded.

WHY IT MATTERS IN QUANT RESEARCH

A chronological split can still leak when long-horizon labels issued before the boundary include returns after it. Purging protects the separation between model fitting and evaluation outcomes.

FORMULA INTUITION

Remove train example i when $(s_i, e_i]$ intersects any held-out label interval or protected test interval under the splitter's policy. The required purge length varies with actual horizons and calendars.

SIMPLE MARKET EXAMPLE

If validation begins July 1 and a training observation from June 27 has a five-session outcome ending July 5, that row must be purged even though its feature timestamp precedes the split.

PYTHON / SYSTEM IMPLEMENTATION

Implement interval intersection using explicit entry and exit timestamps, not a fixed row count. Unit-test mixed horizons, holidays, assets with different calendars, and event labels with variable end times.

CONFUSIONS TO AVOID

Dropping only the final q rows before a split fails with irregular calendars and path-dependent exits. Purging also does not address dependence from features or serial regimes by itself.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The validation service queries label intervals and constructs auditable fold membership with reason codes. Experiment reports show raw, purged, embargoed, and scored counts for every fold.

RELATED CONCEPTS AND QUICK RECALL

Related: information interval, temporal split, interval intersection, fold audit. Recall task: Determine whether a June 27 five-session label belongs in training when validation starts July 1.

Embargo after evaluation windows

INTERMEDIATE

ONE-SENTENCE MEANING

An embargo excludes observations immediately after a held-out period to reduce leakage from lingering information, overlapping features, or sequential dependence. It complements purging but addresses a different direction around the boundary.

MEMORY JOGGER

Purging removes labels that reach into test; embargo protects a buffer after test. Set its length from information persistence, not from a decorative percentage.

WHY IT MATTERS IN QUANT RESEARCH

Feature windows, model state, market impact, and event clustering can carry test-period information into nearby training observations in repeated backtests. A justified embargo makes folds less optimistic at the cost of sample size.

FORMULA INTUITION

For held-out interval ending T , exclude training decisions in $(T, T+b]$ where b is the embargo. In combinatorial or repeated schemes, apply the rule relative to every test block.

SIMPLE MARKET EXAMPLE

A twenty-day rolling feature computed on the first day after a test block contains nineteen test-period observations. An embargo or fold-local state reset prevents that contamination when later data is reused for training.

PYTHON / SYSTEM IMPLEMENTATION

Document the dependency that motivates b , calculate actual calendar boundaries, and report sensitivity to longer buffers. Refit stateful transformations within each fold rather than relying on embargo alone.

CONFUSIONS TO AVOID

Embargo is not a substitute for causal features, and an arbitrary one-percent buffer has no universal meaning. Excessive embargo can make a small dataset unstable and hide rather than quantify dependence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The splitter combines purge, embargo, group, and maturity rules; the feature pipeline respects fold-local fitting. Governance stores boundary parameters as part of experiment identity.

RELATED CONCEPTS AND QUICK RECALL

Related: temporal buffer, feature lookahead, state reset, combinatorial validation. Recall task: Contrast purging with embargo and give one feature-based reason for an embargo after a test block.

Effective sample size and block uncertainty

INTERMEDIATE

ONE-SENTENCE MEANING

Effective sample size is the number of independent-equivalent observations after accounting for serial dependence, overlap, and clustering. It is usually smaller than the label row count and should govern uncertainty claims.

MEMORY JOGGER

Count decisions for operations, but count independent information for statistics. Use blocks long enough to preserve the dependence structure relevant to the target.

WHY IT MATTERS IN QUANT RESEARCH

Naive standard errors and confidence intervals become too narrow when overlapping labels or repeated asset events are treated as independent. Block bootstrap and cluster-robust methods provide more honest uncertainty under stated assumptions.

FORMULA INTUITION

For stationary autocorrelation ρ_k , n_{eff} is approximately $n / (1 + 2 \sum_k \rho_k)$. The approximation can fail under nonstationarity, cross-sectional dependence, and unequal clusters.

SIMPLE MARKET EXAMPLE

One thousand daily five-day labels may contain far less than one thousand independent outcomes. A crash affecting every asset on one date also creates a shared time cluster across the cross-section.

PYTHON / SYSTEM IMPLEMENTATION

Estimate autocorrelation and overlap structure, choose time and entity clusters, and compare several block lengths. Report raw n , cluster counts, effective n , and uncertainty method beside performance metrics.

CONFUSIONS TO AVOID

Simply dividing n by the horizon ignores autocorrelation outside the label window and cross-asset common shocks. Very short blocks break dependence, while very long blocks yield too few resamples.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The statistics layer consumes label intervals and grouping metadata; experiment reports use dependence-aware intervals; model comparison requires enough distinct regimes and event clusters.

RELATED CONCEPTS AND QUICK RECALL

Related: cluster robust error, block bootstrap, autocorrelation, design effect. Recall task: Explain why one thousand overlapping labels can support much less than one thousand independent observations.

Label leakage through future transformations

INTERMEDIATE

ONE-SENTENCE MEANING

Label leakage occurs when target information enters features, sampling, preprocessing, thresholds, or model selection before evaluation. Leakage can be direct, such as a future return column, or indirect, such as full-sample normalization influenced by future outcomes.

MEMORY JOGGER

Trace every dependency backward from the training row. If changing data after the decision can change any model input or sample membership, the pipeline is not causal.

WHY IT MATTERS IN QUANT RESEARCH

Leakage often produces stable-looking performance because the contaminated transformation is applied consistently across folds. Ordinary chronological splits do not repair labels or thresholds constructed with future information.

FORMULA INTUITION

Causal requirement: X_t and sample_t are functions only of data with availability $\leq$ decision t and parameters fitted inside the training fold. Label y_t is isolated from all upstream construction except evaluation.

SIMPLE MARKET EXAMPLE

A three-way label uses a volatility threshold computed from the future outcome window, then that threshold is also passed as a feature. The model receives a compressed description of the answer even though no literal future return column appears.

PYTHON / SYSTEM IMPLEMENTATION

Maintain a dependency graph with temporal contracts, run future-perturbation tests, and fit transformations within folds. Red-team joins, universes, imputation, event triggers, and label-derived sample weights.

CONFUSIONS TO AVOID

Dropping obviously named target columns is insufficient. Target encoding, global class balancing, future-informed universe filters, and post-outcome event definitions can leak without suspicious column names.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The pipeline compiler enforces clock and fold boundaries; lineage tools identify paths from outcomes to inputs; validation gates reject any artifact with an unresolved future dependency.

RELATED CONCEPTS AND QUICK RECALL

Related: target leakage, post-treatment variable, fold isolation, perturbation test. Recall task: Describe an indirect leakage path involving a label threshold and propose a test that would expose it.

Label tuning and multiple testing

INTERMEDIATE

ONE-SENTENCE MEANING

Changing horizon, benchmark, threshold, barrier, scaling, or sample rule after viewing results is a search over target definitions. The winning label can overfit noise even when the predictive model itself is simple.

MEMORY JOGGER

Treat target families as hypotheses with a budget. Select inside training or inner validation and evaluate the complete selection procedure on untouched outer data.

WHY IT MATTERS IN QUANT RESEARCH

Researchers often report label choice as preprocessing, hiding a large source of researcher degrees of freedom. Search breadth should enter uncertainty and governance just like model hyperparameters.

FORMULA INTUITION

Observed best score $\max_j S_j$ is upward biased when J target variants are tried. Correlation among variants reduces but does not eliminate the effective number of trials.

SIMPLE MARKET EXAMPLE

Testing 1-, 2-, 5-, 10-, and 20-day returns plus three cost thresholds creates fifteen related objectives. Selecting the best test Sharpe and presenting it as predetermined exaggerates evidence.

PYTHON / SYSTEM IMPLEMENTATION

Pre-register target rationale, log every variant, use nested temporal validation, and retain specification curves. Promote only if nearby reasonable definitions support the same economic conclusion.

CONFUSIONS TO AVOID

A label with the highest predictive metric may be the least implementable because of cost, maturity delay, or turnover. Correcting only for model hyperparameters while ignoring target search understates selection risk.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment registry records target lineage and trial families; the validation service nests selection; review reports show all attempted definitions and outer-period performance.

RELATED CONCEPTS AND QUICK RECALL

Related: researcher degrees of freedom, specification curve, nested validation, trial budget. Recall task: Count the target variants in a grid of five horizons and three thresholds and explain how to evaluate the selection honestly.

Class imbalance and prevalence drift

INTERMEDIATE

ONE-SENTENCE MEANING

Class imbalance means some label outcomes are much rarer than others, while prevalence drift means those class shares change over time or cohorts. Both affect training losses, probability calibration, and the meaning of headline metrics.

MEMORY JOGGER

A rare event is not a data error. Preserve natural base rates in evaluation and make any sampling or weighting correction explicit.

WHY IT MATTERS IN QUANT RESEARCH

Crash, barrier-hit, and fill labels can have extreme imbalance, making accuracy trivial and unstable. Resampling may help optimization, but predicted probabilities must be corrected or recalibrated for deployment prevalence.

FORMULA INTUITION

prevalence $\pi = P(y=1)$. Under case-control sampling, posterior odds require adjustment by deployment versus sampled class odds before probabilities are interpreted economically.

SIMPLE MARKET EXAMPLE

If only 2% of observations are upper-barrier hits, an always-negative classifier is 98% accurate and useless. A model with 30% precision may still be valuable if recall and payoff on rare events justify action.

PYTHON / SYSTEM IMPLEMENTATION

Report confusion matrices, precision-recall curves, calibration by time and cohort, and expected utility. Fit weights inside training folds and evaluate on untouched data retaining natural prevalence.

CONFUSIONS TO AVOID

Randomly oversampling positives does not create new market regimes. ROC AUC can look strong while precision remains poor at realistic base rates, and a fixed threshold can fail when prevalence shifts.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Training handles optimization weights; calibration restores deployment probabilities; policy chooses thresholds from utility; monitoring tracks class rate, score distribution, and conditional payoff.

RELATED CONCEPTS AND QUICK RECALL

Related: base rate, precision-recall, case-control sampling, calibration. Recall task: Explain why 98% accuracy can be useless when positives occur 2% of the time and name a better evaluation view.

Noisy labels and ambiguity zones

INTERMEDIATE

ONE-SENTENCE MEANING

A noisy label differs from the latent economic outcome because of measurement error, arbitrary boundaries, asynchronous prices, or inherently ambiguous paths. Noise can be random, feature-dependent, or regime-dependent, with very different consequences.

MEMORY JOGGER

Do not pretend a five-basis-point outcome is known precisely when spread and timestamp uncertainty are larger. Expose ambiguity and test whether the model learns the measurement process.

WHY IT MATTERS IN QUANT RESEARCH

Label noise limits achievable performance and can make complex models chase endpoint artifacts. Confidence weights, neutral zones, probabilistic targets, or repeated measurements can be better than hard corrections.

FORMULA INTUITION

Observed $y_t = y^*_t + \epsilon_t$ for continuous outcomes; for classes, $P(y_{\text{obs}} \neq y_{\text{true}} | x)$ may vary with liquidity and volatility. Feature-dependent noise can bias the learned boundary.

SIMPLE MARKET EXAMPLE

A thin stock's official close differs from the executable bid by 25 basis points, while its target threshold is 10 basis points. The class may represent closing-print mechanics more than attainable direction.

PYTHON / SYSTEM IMPLEMENTATION

Store source quality, spread, timestamp tolerance, path ambiguity, and alternative price labels. Compare model conclusions under robust definitions and downweight only with causal, validated quality signals.

CONFUSIONS TO AVOID

Smoothing every label removes genuine tail events, while hand-correcting surprising outcomes invites hindsight. Confidence weights computed from future path quality can leak the answer.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Data quality produces uncertainty metadata; label generation retains alternative views; training may use soft labels or weights; governance reviews sensitivity before promotion.

RELATED CONCEPTS AND QUICK RECALL

Related: measurement error, soft label, ambiguity, executable price. Recall task: Give an example where the bid-ask spread is wider than the class threshold and explain why the hard label is questionable.

Sample weights and economic importance

INTERMEDIATE

ONE-SENTENCE MEANING

Sample weights determine how strongly each labeled observation influences fitting or evaluation. They can represent payoff magnitude, uniqueness, confidence, recency, exposure, or policy value, but every weighting scheme changes the estimand.

MEMORY JOGGER

Weight because the decision values observations differently, not because a backtest improves. Distinguish training optimization weights from evaluation and capital weights.

WHY IT MATTERS IN QUANT RESEARCH

Overlapping events, duplicate information, rare costly errors, and noisy prices may justify nonuniform influence. Poorly designed weights can amplify outliers, leak outcomes, or make metrics impossible to compare.

FORMULA INTUITION

Weighted loss = $\sum_i w_i L(y_i, f_i) / \sum_i w_i$. Uniqueness weights often decrease when many label intervals share the same underlying outcome events.

SIMPLE MARKET EXAMPLE

Five overlapping labels surrounding one earnings shock should not count like five unrelated shocks. Assigning lower uniqueness weight reduces repeated influence while preserving the time sequence.

PYTHON / SYSTEM IMPLEMENTATION

Compute weights from causal metadata, normalize within training folds, cap extremes, and log effective weight concentration. Always report unweighted results and sensitivity to plausible alternatives.

CONFUSIONS TO AVOID

Weighting by absolute realized future return can be useful for payoff emphasis but is outcome-dependent and changes calibration. Using final portfolio exposure as a training weight can also create a circular policy target.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label table carries candidate metadata; training configuration selects a versioned weight policy; evaluation separates statistical, economic, and capital-weighted views.

RELATED CONCEPTS AND QUICK RECALL

Related: uniqueness weight, importance sampling, heteroskedastic loss, weighted metric. Recall task: Propose a uniqueness weight for overlapping event labels and state what unweighted result should still be reported.

Target-specific metrics and calibration

INTERMEDIATE

ONE-SENTENCE MEANING

Metrics must match target type and decision cost; calibration asks whether forecast magnitudes or probabilities correspond to realized outcomes, not only whether their ordering is correct.

MEMORY JOGGER

Choose a primary metric before seeing results and support it with diagnostic metrics. A single score cannot reveal shape, tails, stability, and monetization.

WHY IT MATTERS IN QUANT RESEARCH

Mean squared error emphasizes large misses, rank correlation ignores scale, log loss penalizes confident probability errors, and concordance handles censored timing. Strategy value still requires a policy-level net return test.

FORMULA INTUITION

Regression calibration can test $E[y|f] = a + b f$ with ideal $a=0, b=1$. Probability calibration compares predicted p with empirical frequency; ranking metrics evaluate order rather than numeric accuracy.

SIMPLE MARKET EXAMPLE

A return model may have modest correlation but excellent top-decile spread, while a direction model may have strong AUC but poor probability calibration. Those models support different sizing rules.

PYTHON / SYSTEM IMPLEMENTATION

Publish metric definitions, sample masks, weights, uncertainty, and cohort slices with every result. Use time-aware calibration plots and decision curves tied to realistic cost and risk assumptions.

CONFUSIONS TO AVOID

Optimizing MSE for a heavy-tailed return label can overemphasize crashes, while optimizing rank correlation can ignore economically tiny spreads. Calibration measured on resampled class prevalence is not deployment calibration.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evaluation service selects metrics from the label schema; policy simulation adds utility; dashboards monitor calibration drift and realized spread by score bucket.

RELATED CONCEPTS AND QUICK RECALL

Related: Brier score, information coefficient, concordance, reliability curve. Recall task: Match one metric to a continuous return, probability, ranking, and censored duration target, then name the policy-level check.

Label pipeline, lineage, and reproducibility

INTERMEDIATE

ONE-SENTENCE MEANING

A production label pipeline converts versioned future evidence into mature outcomes with full lineage across source snapshots, code, configuration, calendars, and dependencies.

MEMORY JOGGER

Every label should answer: which prediction, which source events, which rules, which vintage, and which builder produced me. Rebuilding from the same identity must yield the same value or a reason-coded revision.

WHY IT MATTERS IN QUANT RESEARCH

Research conclusions often become irreproducible when vendors rewrite history, benchmark membership changes, or corporate actions are repaired. Immutable artifacts preserve the evidence used for model approval.

FORMULA INTUITION

`label_artifact_hash = H(contract, code, source_snapshot, calendar, reference_versions, dependencies)`. Row lineage maps prediction ID to entry, exit, path, and adjustment events.

SIMPLE MARKET EXAMPLE

A split correction changes historical prices months after an experiment. The old artifact remains auditable, while a new label version quantifies which rows and metrics changed before any model is retrained.

PYTHON / SYSTEM IMPLEMENTATION

Use idempotent jobs, immutable partitions, content hashes, schema tests, and row-level lineage. Emit build manifests with counts by maturity, exclusion reason, revision, horizon, and cohort.

CONFUSIONS TO AVOID

A mutable 'latest' target table destroys experiment identity. Recomputing labels without pinning calendars or benchmark constituents can change outcomes even when price files are unchanged.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The orchestrator matures labels, the registry catalogs artifacts, experiments reference hashes, and governance compares revisions. Incident tools trace a source correction to affected models and decisions.

RELATED CONCEPTS AND QUICK RECALL

Related: data lineage, artifact hash, idempotence, immutable snapshot. Recall task: List the components of a label artifact hash and explain why a source-vendor correction should create a new artifact.

Delayed feedback and live performance monitoring

INTERMEDIATE

ONE-SENTENCE MEANING

Forecasts cannot be scored until their targets mature, creating a feedback delay that grows with horizon and revision policy. Live monitoring must separate input health, forecast behavior, provisional outcomes, and final target performance.

MEMORY JOGGER

Do not interpret missing recent labels as good or bad performance. Track cohorts by issue date and show how much outcome exposure has matured.

WHY IT MATTERS IN QUANT RESEARCH

A twenty-day model may trade for a month before a complete performance cohort exists. Operational alarms therefore need leading indicators, while model-quality decisions require enough mature, dependence-aware evidence.

FORMULA INTUITION

maturity coverage at time T = mature predictions issued in cohort / eligible predictions issued in cohort. Score aggregation must avoid mixing partial and fully mature horizons without masks.

SIMPLE MARKET EXAMPLE

The newest weekly dashboard contains only one-day labels but almost no twenty-day labels. Comparing its raw average to older complete weeks would produce a false recent deterioration or improvement.

PYTHON / SYSTEM IMPLEMENTATION

Build cohort tables with issued, entered, mature, revised, and excluded counts. Provide provisional mark-to-market separately from contract-final outcomes and delay automated retraining until evidence thresholds are met.

CONFUSIONS TO AVOID

Backfilling late labels into a current-period chart can make history move without explanation. Early-maturing observations may differ systematically from halted or complex cases, producing maturity selection bias.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Operations watches data and score health immediately; evaluation updates as outcomes mature; governance uses minimum cohort and regime requirements for action. Reports label every metric as provisional or final.

RELATED CONCEPTS AND QUICK RECALL

Related: feedback delay, cohort analysis, provisional metric, maturity coverage. Recall task: Explain why the newest cohort of a twenty-day model cannot be compared directly with a fully mature historical cohort.

Target drift, governance, and change control

INTERMEDIATE

ONE-SENTENCE MEANING

Target drift occurs when the distribution, prevalence, noise, or economic relationship of outcomes changes, even if the label code is unchanged. Change control distinguishes market drift from deliberate contract revisions and data corrections.

MEMORY JOGGER

Monitor what the model is asked to predict as carefully as its features. Any change in horizon, price source, benchmark, cost, universe, or maturity policy is a governed target migration.

WHY IT MATTERS IN QUANT RESEARCH

Volatility regimes change return scale, market structure changes fill outcomes, and benchmark reconstitutions change relative labels. Uncontrolled target drift can break calibration and make old training weights inappropriate.

FORMULA INTUITION

Track $P_t(y)$, conditional $P_t(y|\text{cohort})$, missing rate, revision rate, and proxy-to-utility mapping. Drift statistics need economic thresholds and stable reference windows rather than automatic retraining triggers alone.

SIMPLE MARKET EXAMPLE

A spread-widening regime doubles the neutral-band crossing rate from 15% to 30%. The label definition is stable, but class prevalence and transaction-cost relevance have changed.

PYTHON / SYSTEM IMPLEMENTATION

Version all contract changes, run parallel rebuilds, compare affected rows and decisions, and require approval for model migration. Monitor distribution, calibration, coverage, revisions, and economic payoff by cohort and regime.

CONFUSIONS TO AVOID

Retraining on corrupt targets can institutionalize a data incident. Treating every natural regime change as a schema change also fragments history and prevents coherent long-run evaluation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label owner, model owner, risk, and data steward share an approval path. Rollback points preserve prior artifacts, and incident reviews update tests, thresholds, and migration playbooks.

RELATED CONCEPTS AND QUICK RECALL

Related: concept drift, schema migration, prevalence shift, rollback. Recall task: Describe a target-distribution change that requires recalibration but not a new label definition, and one change that does require a new version.

Point-in-time target pipeline

INTERMEDIATE

1

Prediction registry

Decision ID, model, cutoff, universe, score, and intended action

2

Execution mapping

Eligible entry event, price convention, latency, side, and size assumption

3

Future evidence

Prices, trades, benchmarks, corporate actions, and path events

4

Outcome resolver

Endpoint, interval, barrier, rank, risk, or survival calculation

5

Quality and maturity

Completeness, censoring, ambiguity, provisional state, and revisions

6

Immutable artifact

Contract, source vintage, lineage, tests, counts, and content hash

7

Validation geometry

Intervals, groups, purge, embargo, weights, and uncertainty

8

Feedback and governance

Cohort metrics, drift, policy value, migration, and rollback

CONTROL PRINCIPLE

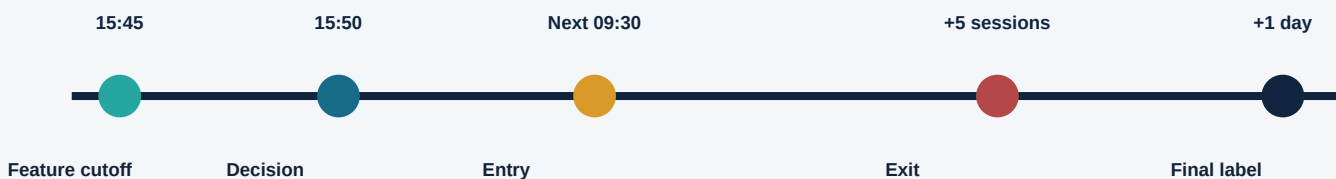
A label is an outcome artifact tied to one prediction decision. Entry, path, exit, quality, maturity, and vintage must remain reconstructable even after later data corrections.

Decision, entry, horizon, and maturity

INTERMEDIATE

SCENARIO

A model uses data frozen before the close, enters in the next session, measures a five-session return, and waits for final corporate-action processing. Each stage has a different clock and owner.



FEATURE BOUNDARY

Only evidence available by 15:45 may enter features or fitted state. The final close is outside the contract even though it shares the same calendar date.

EXECUTION BOUNDARY

The 15:50 score cannot claim the next session's open in advance. The label entry is the modeled fill event, including auction eligibility and any fallback rule.

OUTCOME INTERVAL

Five sessions are measured from the executable entry, not from the feature cutoff. Holidays, halts, and missing exits are resolved by the calendar policy.

MATURITY BOUNDARY

The endpoint exists after session five, but final scoring waits for the declared revision lag. Until then, the outcome is provisional rather than missing.

AUDIT RECORD

Store every timestamp, price event, source vintage, and reason code so a later correction creates a traceable revision instead of an unexplained historical change.

Target formula comparison

INTERMEDIATE

TARGET	CORE FORM	BEST MATCH	PRIMARY CAUTION
Simple return	$P_{\text{exit}}/P_{\text{entry}} - 1$	Wealth outcome	Must compound
Log return	$\log(P_{\text{exit}}/P_{\text{entry}})$	Additive time algebra	Large-move interpretation
Excess return	$R_{\text{asset}} - R_{\text{bench}}$	Relative mandate	Aligned benchmark
Risk-scaled	$R / \sigma_{\text{prior}}$	Cross-asset comparison	Causal denominator
Direction	$1[R_{\text{net}} > 0]$	Directional action	Magnitude discarded
Three-way	sign outside band	Trade or abstain	Threshold search
Rank	percentile in U_t	Long-short ordering	Peer-set bias
Barrier	first boundary hit	Stop-profit-time policy	Within-bar order
Realized vol	$\sqrt{\text{sum } r_u^2}$	Risk forecast	Sampling noise

READING RULE

The visible equation is only one part of the target. Entry and exit clocks, executable price, adjustments, universe, maturity, missing policy, and source vintage complete the definition.

Endpoint return versus path outcomes

INTERMEDIATE

SCENARIO

Two synthetic five-session paths start at 100 and end at 102. Their endpoint return is identical, but drawdown, barrier order, and realizable exit behavior differ materially.



INTERPRETATION

Both endpoint labels are +2%. The volatile path first breaches the lower barrier, later reaches the upper barrier, and carries larger MAE and MFE; a path-aware strategy would not treat the examples as equivalent.

IMPLEMENTATION NOTE

Bar data may show both barriers without revealing order. Use higher-frequency evidence, a conservative tie rule, and an ambiguity flag rather than inventing a favorable sequence.

Worked example: triple-barrier label

INTERMEDIATE

CONTRACT

A long enters at 100 after the decision. Prior volatility sets an upper barrier at 103, a lower barrier at 97, and a vertical barrier after five sessions.

OBSERVED PATH

The ordered closes are 100.8, 98.5, 96.7, 101.2, and 104.0. The third close crosses the lower barrier before the later recovery.

FIRST-HIT RESULT

The class is -1 with first-hit time equal to session three. The later upper crossing does not revise the event because the trading rule would already have exited.

GAP HANDLING

If the market opens at 95, record the actual executable crossing or a conservative fill model rather than assuming a fill exactly at 97.

WITHIN-BAR AMBIGUITY

If one daily bar has high 104 and low 96, the daily record cannot identify which barrier came first. Use intraday events or emit an ambiguity state.

CAUSAL SCALE

The 3% distance must come from volatility or risk known before entry. A multiplier calibrated from this future path would leak the answer.

VALIDATION INTERVAL

The label interval ends at the first valid hit, not automatically at session five. Purging should use the actual event end for this observation.

AUDIT FIELDS

Store entry, barriers, source events, first-hit price and time, side, tie rule, maturity, source vintage, and label version.

POLICY CHECK

A predictive improvement matters only if the simulator can trade the same barrier rule after spread, slippage, gap risk, and capacity.

How forward labels overlap

INTERMEDIATE

FIVE-SESSION TARGET ISSUED DAILY

Each row below consumes five future sessions. Adjacent labels share four sessions, so one realized shock enters several labels and naive row-level uncertainty becomes too optimistic.

	D1	D2	D3	D4	D5	D6	D7	D8
label D1		used	used	used	used	used		
label D2			used	used	used	used	used	
label D3				used	used	used	used	used
label D4					used	used	used	used

DEPENDENCE

Label D1 and label D2 share sessions D3 through D6. Under independent daily returns, adjacent five-session sums have approximate correlation $4/5 = 0.8$ before any additional market persistence.

VALIDATION CONSEQUENCE

A split boundary must purge training labels whose intervals touch held-out outcomes. Uncertainty should use blocks or event clusters, and reports should distinguish raw rows from effective evidence.

ALTERNATIVE

Thinning to non-overlapping labels is simple but discards data and depends on the chosen phase. Interval-aware methods retain observations while acknowledging dependence and protecting fold boundaries.

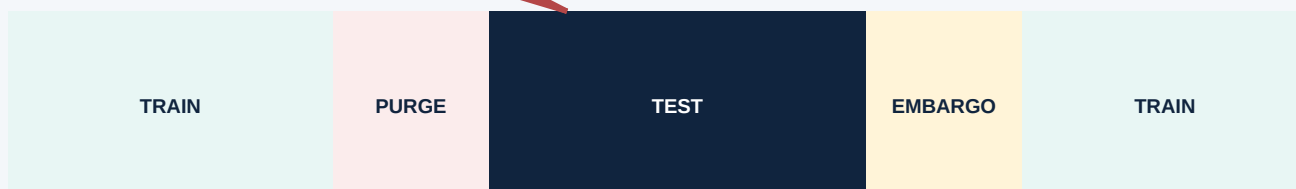
Purging and embargo around a test block

INTERMEDIATE

BOUNDARY PROBLEM

Feature timestamps can sit safely before a split while their outcome intervals cross into test. Observations immediately after test may also carry test-period information through rolling state or event persistence.

training labels whose future intervals enter test



PURGING

Remove training examples whose label intervals intersect the test or validation outcome region. Calculate actual timestamp intersections for mixed and variable horizons.

EMBARGO

Exclude a justified buffer after the held-out block when rolling state, event clusters, or sequential dependence can transmit test information into later training rows.

FOLD-LOCAL FITTING

Refit scalers, thresholds, encoders, volatility estimates, and stateful transforms within the training fold. Boundary buffers do not legalize full-sample parameters.

AUDIT COUNTS

Report original, purged, embargoed, immature, censored, and scored counts. Large losses in usable data are evidence about the true information content.

SENSITIVITY

Repeat with longer buffers and alternative block designs. A result that vanishes under a modest defensible change is not operationally robust.

Decision tree for choosing a target

INTERMEDIATE

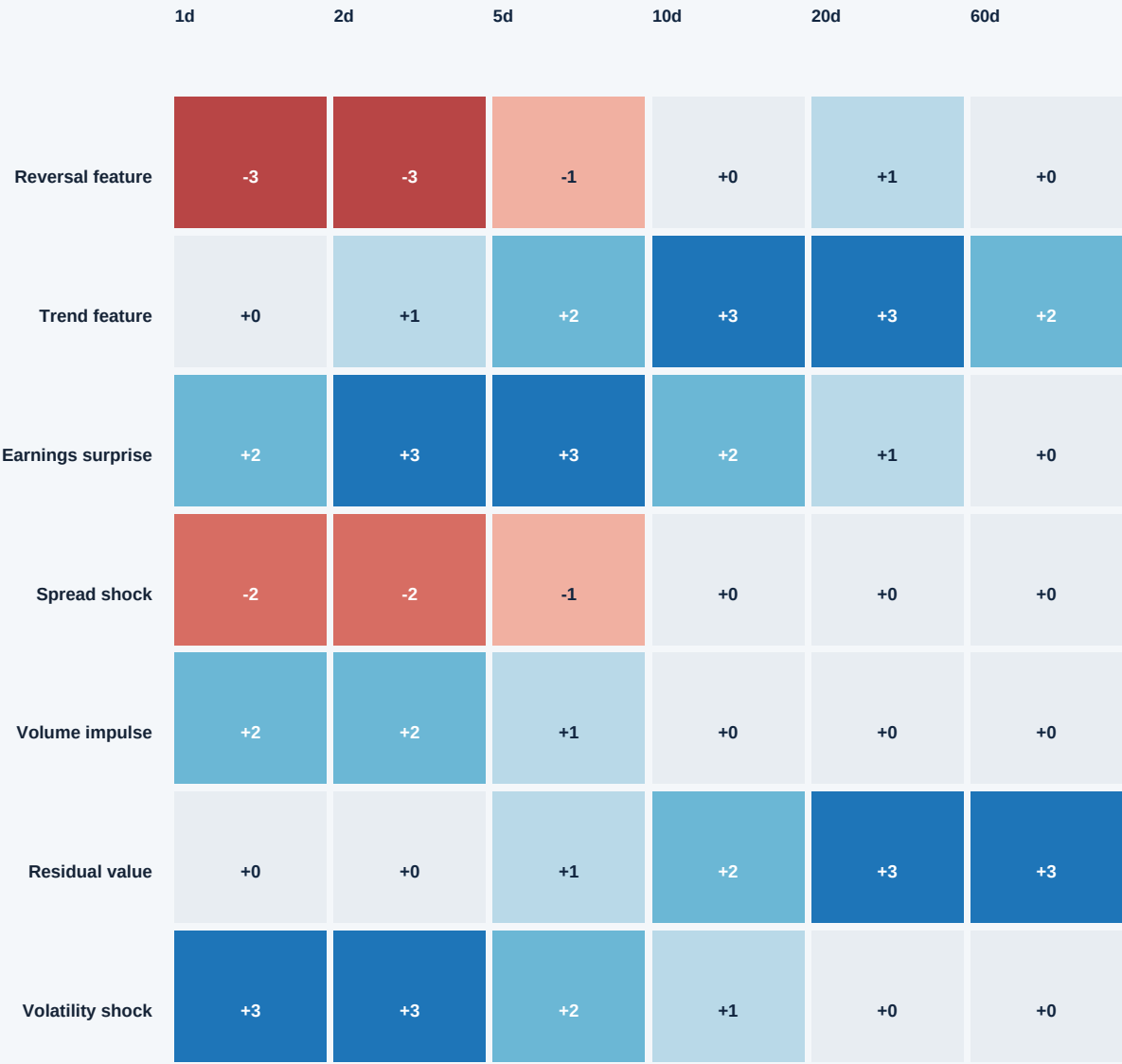


SELECTION RULE

Choose from decision mechanics first, then validate learnability and economics. A target is not correct merely because one model scores well on it.

Multi-horizon target behavior

INTERMEDIATE



INTERPRETATION

The synthetic map shows horizon-dependent sign and decay. Short reversal can fade or reverse, while trend and value may need longer windows; averaging all horizons into one score would hide this structure.

VALIDATION STANDARD

Define the horizon family before final evaluation, fit any selection inside inner folds, and report the full curve with dependence-aware uncertainty. Portfolio design should match score decay rather than automatically hold for the winning target horizon.

Rare-event labels and confusion economics

INTERMEDIATE

EXAMPLE

Upper-barrier hits occur in 2% of 10,000 observations. A model predicts 300 positives, correctly finding 120 of the 200 actual events and falsely flagging 180 ordinary cases.

	ACTUAL HIT	ACTUAL NO HIT
PREDICT HIT	120	180
PREDICT NO HIT	80	9620

ACCURACY: 97.4%

Dominated by the majority class

RECALL: 60.0%

120 of 200 actual hits found

PRECISION: 40.0%

120 of 300 predicted hits are real

BASE RATE: 2.0%

Sets the difficulty of useful precision

ECONOMIC DECISION

Precision and recall still do not settle deployment. Attach average profit on true hits, loss on false positives, missed-opportunity cost, execution cost, and probability calibration before choosing the action threshold.

Worked calculation: return, excess, and risk units

INTERMEDIATE

CONTRACT

Enter at the next open of 100 and exit five sessions later at 103. Prices are total-return adjusted, the benchmark moves from 200 to 204, and prior five-day volatility forecast is 1.5%.

SIMPLE RETURN

Asset simple return is $103/100 - 1 = 3.00\%$. This is the direct wealth change before transaction costs.

LOG RETURN

Asset log return is $\log(1.03) = 2.956\%$. It differs slightly from simple return and is additive across subperiods.

BENCHMARK RETURN

Benchmark return is $204/200 - 1 = 2.00\%$. The two legs use aligned entry and exit sessions under the same currency convention.

EXCESS RETURN

Arithmetic excess return is $3.00\% - 2.00\% = 1.00\%$. This is the active outcome for a benchmark-relative mandate.

RISK-SCALED RETURN

Ex-ante standardized outcome is $3.00\% / 1.50\% = 2.0$ risk units. The denominator was frozen before entry and uses a documented floor.

NET TARGET

If round-trip cost is 0.20%, net asset return is 2.80%. Whether benchmark cost is subtracted depends on whether the strategy explicitly trades the hedge.

CLASSIFICATION VIEWS

Direction is positive; a +/-0.50% neutral band also yields positive; cross-sectional rank requires outcomes for the full eligible peer set.

AUDIT FIELDS

Store entry and exit events, asset and benchmark components, volatility artifact, cost assumption, raw and transformed targets, maturity, and label version.

Target failure modes and first checks

INTERMEDIATE

SYMPTOM	LIKELY CAUSE	FIRST CHECK	ACTION
Same-close alpha	Impossible entry	Feature and fill clocks	Move entry or freeze earlier
Test result too smooth	Overlapping labels	Interval map	Block inference and purge
Recent metric missing	Immature horizon	Cohort maturity	Wait or mark provisional
High accuracy, poor PnL	Magnitude ignored	Payoff matrix	Use utility and calibration
Rank result unstable	Peer-set drift	Universe snapshot	Repair point-in-time group
Barrier labels flip	Within-bar ambiguity	Higher-frequency path	Tie rule and flag
Targets change on rebuild	Mutable source history	Artifact manifest	Pin vintage and version
Rare class collapses	Prevalence shift	Rate by cohort	Recalibrate threshold
Live edge below research	Proxy-policy gap	Cost and fill attribution	Revise policy or reject

TRIAGE ORDER

Freeze the label artifact, preserve source vintages, identify the first divergent decision ID, quantify affected folds and models, rebuild in parallel, then publish a governed superseding version.

Maturity cohorts and delayed feedback

INTERMEDIATE

	W-5	W-4	W-3	W-2	W-1	NOW
1-day	FINAL	FINAL	FINAL	FINAL	FINAL	WAIT
5-day	FINAL	FINAL	FINAL	FINAL	PART	WAIT
20-day	FINAL	FINAL	PART	PART	WAIT	WAIT

WHY THE TRIANGLE APPEARS

Longer horizons need more future time. Recent issue cohorts contain valid predictions but do not yet contain complete outcomes, so their absence must not be treated as zero performance.

PARTIAL OUTCOMES

Mark-to-market paths may support operational awareness, but they are not contract-final targets. Keep provisional and final metric series visually and logically separate.

FAIR COMPARISON

Compare cohorts only at the same maturity age or wait for completion. Otherwise recent dashboards overrepresent short horizons and easy-to-resolve cases.

ACTION THRESHOLD

Automated retraining or rollback should require minimum mature predictions, distinct event clusters, and regime coverage. Clock time alone is not enough evidence.

REVISION VISIBILITY

When final labels replace provisional values, publish the revision count and metric delta. Historical charts should explain movement rather than silently rewriting it.

Target promotion checklist

INTERMEDIATE

DECISION AND UTILITY

Actor, cutoff, executable entry, horizon, policy use, payoff asymmetry, costs, and risk objective are explicit.

OUTCOME CONTRACT

Formula, path rule, units, benchmark, universe, price convention, adjustments, calendar, maturity, missingness, and censoring are versioned.

CAUSAL BOUNDARIES

Features stop before decision, entry follows feasible execution, thresholds and scales are fitted in training, and sample membership does not use future outcomes.

DATA QUALITY

Entry and exit coverage, delistings, halts, bad ticks, stale references, within-bar ambiguity, revisions, and cohort selection have been quantified.

VALIDATION GEOMETRY

Every label interval is available to the splitter; purge, embargo, grouping, overlap, effective sample size, and block uncertainty are justified.

TARGET SEARCH

All horizons, bands, barriers, benchmarks, and weight schemes are logged; selection occurs inside inner validation and nearby specifications are reported.

METRIC FIT

Primary statistical metric matches target type, probabilities or magnitudes are calibrated, and policy-level net utility survives realistic execution.

REPRODUCIBILITY

Source snapshot, calendars, reference data, code, dependencies, row lineage, tests, artifact hash, owner, and supersession policy are complete.

OPERATIONS

Maturity cohorts, provisional reporting, revisions, drift, prevalence, calibration, incident scope, migration, and rollback actions are monitored and rehearsed.

Final active recall and system index

INTERMEDIATE

TARGET MEANING

State what future object is predicted, its units, and why that object supports the intended decision rather than merely producing a convenient metric.

CLOCK CHAIN

Trace feature cutoff, decision, executable entry, outcome interval, exit, maturity, and revision. Explain why no future event can influence sample construction.

OUTCOME FAMILY

Choose among raw, excess, standardized, rank, direction, barrier, excursion, risk, and survival labels from strategy mechanics.

INCOMPLETE OUTCOMES

Distinguish immature, missing, censored, delisted, ambiguous, revised, and excluded states. Never collapse them to an unexplained zero.

INTERVAL GEOMETRY

Draw overlap and split boundaries, then justify purge, embargo, grouping, block length, and effective evidence.

SEARCH DISCIPLINE

Count every horizon, threshold, benchmark, barrier, sampling, and weighting variant as part of the research search.

METRIC DISCIPLINE

Match metrics to the label, report calibration and uncertainty, and verify that the downstream policy converts forecasts into positive net utility.

ARTIFACT DISCIPLINE

Bind contract, source vintage, reference data, code, tests, row lineage, maturity, and revision history to one immutable label identity.

FINAL RECALL

For any model result, reconstruct one prediction's features, entry, outcome path, final target, fold membership, sample weight, metric contribution, and production status.